

Core Auth Regression Test Solution

AI Generation Output Validation

Current foundation: **ATL105**

End goal: generate and validate test artifacts for any specification.

1. Business Problem

AI artifacts can be valid JSON while still being incomplete, untraceable, semantically wrong, or based on unapproved knowledge.



The Test Solution is the independent quality gate.

2. AI Solution Outcomes

The AI Solution provides outcome artifacts for Test Validation:

AI outcome	What it contains	Validation focus
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Knowledge model	Fields, rules, values, relationships, source references	Provenance, confidence, flags, approvals
Business Requirements	Testable source-derived statements	Source mapping, completeness, atomicity
Test Scenarios	Business situations to test	Relevance, classification, missing scenarios
Test Cases	Objectives, conditions, expected results, links	Traceability, consistency, approval state
Test Data JSON	Requests, values, lifecycle data	Schema, semantics, serialization
Traceability/coverage	Artifact links and reported metrics	Independent recalculation

3. Test Validation Stages

AI outcome artifacts	→	Preserve	→	Normalize
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Validate structures	→	Provenance/approvals	→	Rule → BR → Scenario → Case → Data
ATL105 semantics	→	Independent coverage	→	High-risk sample and report

The Test Solution validates AI outcomes. It does not need to reproduce the AI generation process to determine whether the outcomes are complete, traceable, and correct.

4. Approval Gates

AI output	→	Preserve	→	Normalize
Validate stages	→	Provenance/approvals	→	Rule → BR → Scenario → Case → Data
ATL105 semantics	→	Independent coverage	→	Decision/report/review

The Test Solution compares AI claims with independent rules and evidence at every stage.

Before broad acceptance, the Test Team manually validates only a small, risk-based handful of high-risk AI artifacts against canonical Test Solution artifacts, independent specification rules, and expected business behavior. This targeted sample confirms compatible results for high-impact items; it does not replace automated validation or formal approval.

4. Approval Gates

Knowledge review	→	Scenario review	→	Test-case review
Source evidence	→	Approved scenario baseline	→	Approved regression suite

Confidence scoring can prioritize attention but cannot replace SME/TBA approval.

5. Traceability

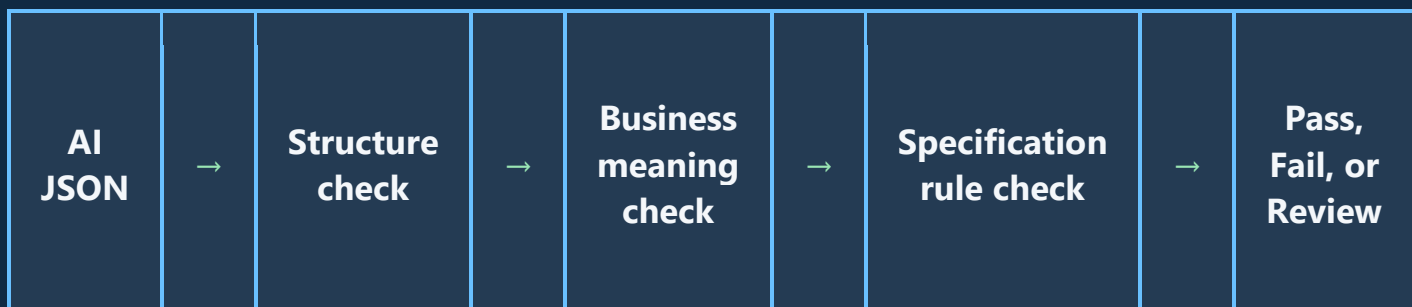
```
Approved source rule
-> Business Requirement
    -> Test Scenario
        -> Test Case
            -> Test Data
                -> Executable payload
```

Canonical Approach



The common format contains the requirement, scenario, test case, test data, source reference, and expected result. The AI original ID is preserved, but validation uses the common structure and source rule.

Semantic Approach



Structural validation: Is the JSON written correctly?

Semantic validation: Does the JSON mean the correct business behavior?

Example: Segment Type 101 can be valid JSON, but it is wrong when the test claims to represent Segment 100.

6. ATL105 Foundation Plan

ATL105 foundation plan

```
-> establish specification profile and source rule
catalog
-> define all ATL105 segments and message categories
-> derive requirements, scenarios, cases, and test data
-> define approvals, traceability, coverage, and
dependency governance
-> expand module by module across the complete ATL105
scope
```

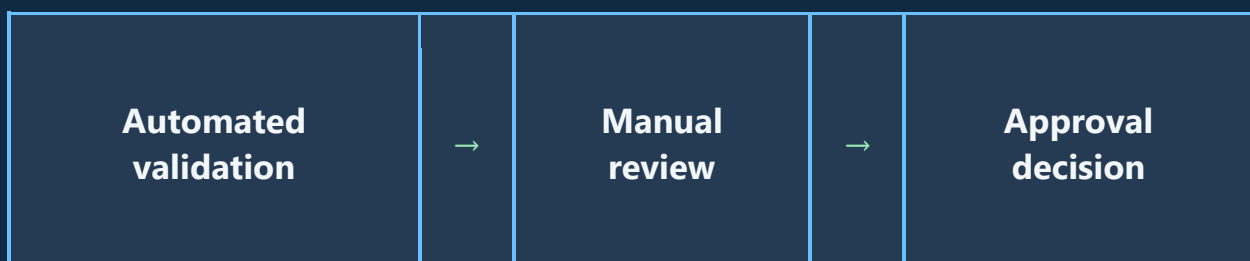
7. Coverage and Reporting

Proposed model - pending Fiserv Leadership approval.

```
Rule Coverage = mandatory rules with complete executable
evidence
/ mandatory rules in the independent rule
catalog x 100
```

Additional metrics: requirement, scenario, test-case, test-data, traceability, semantic-execution, parameter, and segment coverage.

8. Governance and Decision



Unknown or ambiguous combinations are held for manual review.

```
APPROVED
APPROVED_WITH_REVIEW_ITEMS
REJECTED_SCHEMA
REJECTED_TRACEABILITY
REJECTED_SEMANTIC_MISMATCH
REJECTED_MISSING_COVERAGE
```

9. RAID Log

Scope: Any available specification

Current proof of concept: ATL105

Category	ID	Dependency	Required For	Owner	Status
Dependency	D-001	Manual validation of a representative AI artifact sample against Test Solution output	Confirm compatible expected results before broader acceptance	Coforge Test Team	Open
Dependency	D-002	Availability of an SME/TBA or delegated business approver	Review and approve knowledge, scenarios, test cases, ambiguous rules, and business certification		Blocked

Allowed statuses: Open, In Progress, Blocked, Accepted, Closed.

10. End Goal

```
Any specification
-> specification pack
-> independent rule catalog
-> generate BRs
-> generate scenarios
-> generate test cases
```

- > generate test data
- > validate AI equivalents
- > coverage and approval report
- > approved regression suite

The Core Auth solution is reusable across specifications; ATL105 is the current foundation.